1. 1. A speech signal is sampled at a rate of 8 kHz, and then encoded using 8 bits per sample. The binary stream is then transmitted through an AWGN baseband channel via M-level PAM (Pulse Amplitude Modulation). The ISI (Inter-Symbol Interference) should be eliminated.

a) Determine the minimum bandwidth required for transmission when $M = 4$.

b) Determine the minimum bandwidth required for transmission when $M = 16$. What is the price to pay for saving the bandwidth?

a) $R_b = 8\,kHz \times 8$ bits per sample $= 64$ kbps

$$W_{min} = \frac{R_b}{2\log_2 M} = \frac{64}{2\log_2 4} = 16\,kHz$$

b) $$W_{min} = \frac{R_b}{2\log_2 M} = \frac{64}{2\log_2 16} = 8\,kHz$$

Saving the bandwidth, M needs increasing.

2. 2. For a 4-level PAM baseband transmission system, the transmitted signals consists of pulses having a raised cosine spectrum with a roll-off factor 0.5 and its spectrum extends to 7.5kHz.

a) What is the maximum data rate with zero-ISI requirement? What is the corresponding maximum bandwidth efficiency?

b) Will there be any ISI when the symbol period is 0.2ms? What about 0.15ms? Explain in the perspectives of both the time domain and the frequency domain.

a) $$\frac{1+\alpha}{2} R_{s,max} = W \Leftrightarrow R_{s,max} = \frac{2W}{1+\alpha} = \frac{2\times 7.5\,kHz}{1+0.5} = 10\,kHz$$

$$\eta_{b,max} = \frac{R_{b,max}}{W} = \frac{R_{s,max}\log_2 M}{W} = \frac{2\log_2 M}{1+\alpha} = \frac{4}{1.5} = \frac{8}{3} \approx 2.67$$

b) Denote the raised cosine spectrum above is $G(f)$

$$G(f) = \begin{cases} \frac{1}{R_{s,max}}, & |f| < \frac{1-\alpha}{2}R_s \\ \frac{1}{2R_{s,max}}\left[1+\cos\left(\frac{\pi}{\alpha R_{s,max}}\left(|f| - \frac{1-\alpha}{2}R_{s,max}\right)\right)\right], & \frac{1-\alpha}{2}R_{s,max} \le |f| \le \frac{1+\alpha}{2}R_{s,max} \\ 0, & |f| > \frac{1+\alpha}{2}R_{s,max} \end{cases}$$

$G(f)$ satisfies : $$\sum_{n=-\infty}^{+\infty} G(f - nR_{s,max}) = \frac{1}{R_{s,max}}, \quad \forall f \in \mathbb{R}$$

(i) $T_s = 0.2ms \Rightarrow R_s = \frac{1}{2}R_{s,max}$

$$\sum_{n=-\infty}^{+\infty} G(f-nR_s) = \sum_{n=-\infty}^{+\infty} G\left(f - \frac{n}{2}R_{s,max}\right) = \sum_{k=-\infty}^{+\infty}\left[G\left(f - \frac{(2k-1)}{2}R_{s,max}\right) + G\left(f - \frac{2k}{2}R_{s,max}\right)\right]$$

$$= \sum_{k=-\infty}^{+\infty}\left[G\left(f + \frac{R_{s,max}}{2} - kR_{s,max}\right) + G(f - kR_{s,max})\right]$$

$$= \frac{2}{R_{s,max}} = R_s$$

$$\Leftrightarrow g(t)\sum_{n=-\infty}^{+\infty}\delta(t - nT_s) = g(t)\sum_{n=-\infty}^{+\infty}\delta\left(t - n\cdot\frac{2}{R_{s,max}}\right) = \delta(t) \quad [\text{No-ISI}]$$

(ii) $T_s = 0.15ms \Leftrightarrow R_s = \frac{2}{3}R_{s,max}$

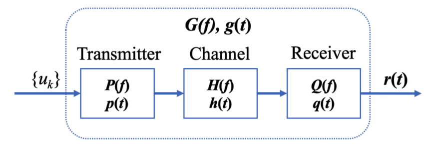
Compute $\left.\sum_{n=-\infty}^{+\infty} G(f-nR_s)\right|_{f=0}$

$$\sum_{n=-\infty}^{+\infty} G(0-nR_s) = \sum_{n=-\infty}^{+\infty} G(nR_s) = G\left(-\tfrac{2}{3}R_{s,max}\right) + G(0) + G\left(\tfrac{2}{3}R_{s,max}\right)$$

$$= \frac{1}{R_{s,max}} + \frac{2}{2R_{s,max}}\left(1+\cos\frac{5\pi}{6}\right) = \frac{2-\frac{\sqrt{3}}{2}}{R_{s,max}} \ne \frac{3/2}{R_{s,max}} = \frac{1}{R_s}$$

$$\Rightarrow \exists\, \{a_k\},\ \sum_{k\ne 0} a_k\, g\left(t - k\cdot\frac{1}{R_s}\right) \ne 0 \quad (\text{Have ISI})$$

3. 3. Consider a PAM baseband transmission system. The baseband waveform is $p(t)$ with the symbol interval T_s, the channel is characterized by a filter $h(t)$, and the filter at the receiver is $q(t)$. As a result, the received waveform is given by $r(t) = \sum_k u_k g(t - kT)$, where $g(t) = p(t) * h(t) * q(t)$ and u_k is the sequence of symbols.



a) What property must $g(t)$ have, so that $r(kT) = u_k$ for any k?

b) Recall the Fourier transform of $\text{sinc}(x)$ is

$$\mathcal{F}\{\text{sinc}(x)\} = \text{rect}(f) = \begin{cases} 1, & f \in [-0.5, 0.5] \\ 0, & \text{else} \end{cases}.$$

Let the transmitter waveform be $p(t) = \text{sinc}(1000t)\text{sinc}(2000t)$. Calculate the frequency domain representation $P(f)$ of the waveform.

c) Assume the symbol time is $T_s = 0.5$ms, and the frequency response of the channel is specified by

$$H(f) = \begin{cases} 1, & |f| \in [0, 0.75\text{kHz}] \\ 0, & |f| \in (0.75\text{kHz}, 1\text{kHz}] \\ 1, & |f| \in (1\text{kHz}, 1.25\text{kHz}] \\ 0, & |f| \in (1.25\text{kHz}, \infty] \end{cases}.$$

Can you design a receiver filter $Q(f)$ so that there is no ISI?

(Hint: The spectrum $G(f)$ of $g(t)$ should satisfy the condition that $G(f) + G(1/T_s - f)$ is a constant, according to the Nyquist criterion)

d) If $H(f)$ is non-zero only at $[0, 0.75\text{kHz}]$, is it possible to design a no-ISI receiver and why?

a) $$g(t)\sum_{k=-\infty}^{+\infty}\delta(t - kT_s) = \delta(t) \quad \left(\text{or } \sum_{n=-\infty}^{+\infty} G\left(f - \frac{n}{T_s}\right) = T_s\right)$$

b) Use Fourier Transform property :

$$X(f) = \int_{-\infty}^{+\infty} x(t)\, e^{-i\cdot 2\pi f t}\, dt = F\{x(t)\}$$

$$F\{x(at)\} = \frac{1}{|a|}X\left(\frac{f}{a}\right)$$

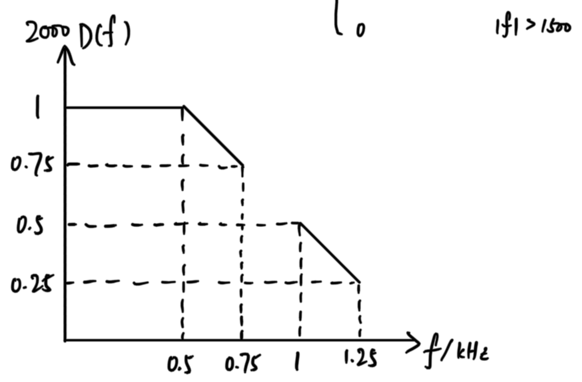
$p(t) = \text{sinc}(1000t)\,\text{sinc}(2000t)$

$$F\{\text{sinc}(1000t)\} = \begin{cases} \frac{1}{1000}, & |f| \le 500 \\ 0, & |f| > 500 \end{cases} \qquad F\{\text{sinc}(2000t)\} = \begin{cases} \frac{1}{2000}, & |f| \le 1000 \\ 0, & |f| > 1000 \end{cases}$$

$$P(f) = F\{\text{sinc}(1000t)\} * F\{\text{sinc}(2000t)\}$$

$$= \frac{1}{1000}\times\frac{1}{2000}\times\begin{cases} 1000\times 1, & |f| \le 500 \\ 1500 - |f|, & 500 < |f| \le 1500 \\ 0, & |f| > 1500 \end{cases}$$

$$= \frac{1}{2000}\times\begin{cases} 1, & |f| \le 500 \\ 1.5 - \left|\frac{f}{1000}\right|, & 500 < |f| \le 1500 \\ 0 & |f| > 1500 \end{cases}$$



c) $D(f) = P(f)\,H(f)$

$$= \frac{1}{2000}\times\begin{cases} 1, & |f| \in [0, 0.5]\text{ kHz} \\ 1.5 - \frac{|f|}{1\text{kHz}}, & |f| \in (0.5, 0.75]\text{ kHz} \\ 0, & |f| \in (0.75, 1]\text{ kHz} \\ 1.5 - \frac{|f|}{1\text{kHz}}, & |f| \in (1, 1.25]\text{ kHz} \\ 0, & |f| \in (1.25, +\infty)\text{ kHz} \end{cases}$$

$$G(f) = D(f)\,Q(f) = \frac{Q(f)}{2000}\times\begin{cases} 1, & |f| \in [0, 0.5]\text{ kHz} \\ 1.5 - \frac{|f|}{1\text{kHz}}, & |f| \in (0.5, 0.75]\text{ kHz} \\ 0, & |f| \in (0.75, 1]\text{ kHz} \\ 1.5 - \frac{|f|}{1\text{kHz}}, & |f| \in (1, 1.25]\text{ kHz} \\ 0, & |f| \in (1.25, +\infty)\text{ kHz} \end{cases} = \frac{Q(f)\,D_1(f)}{2000}, \quad T_s = 0.5ms \Leftrightarrow R_s = \frac{1}{T_s} = 2kHz$$

$$\frac{1}{2000} = \sum_{n=-\infty}^{+\infty} G(f - nR_s) \Leftrightarrow 1 = \sum_{n=-\infty}^{+\infty} D_1(f-nR_s)\,Q(f-nR_s) = \sum_{n=-\infty}^{+\infty} D_1(f-2n)\,Q(f-2n)$$

When $f \in [0, 0.5]$ kHz, $1 = D_1(f)\,Q(f) = Q(f)$

When $f \in (0.5, 0.75)$ kHz, $1 = D_1(f)\,Q(f) = Q(f)\,(1.5 - |f|) \Leftrightarrow Q(f) = \frac{1}{1.5-|f|}$

When $f = 0.75$ kHz, $1 = D_1(0.75)\,Q(0.75) + D_1(-1.25)\,Q(-1.25)$

$= Q(0.75)\cdot 0.75 + Q(-1.25)\cdot 0.25$

When $f \in (0.75, 1)$ kHz, $1 = D_1(f)\,Q(f) + D_1(f-2)\,Q(f-2)$

$= D_1(f-2)\,Q(f-2)$

$= Q(f-2)\,[1.5 - |f-2|] \Rightarrow Q(f-2) = \frac{1}{1.5-|f-2|}$

When $f = 1$kHz, $1 = D_1(1)\,Q(1) + D_1(-1)\,Q(-1)$

$= 0$, 矛盾 $[Q(f)\text{不存在}]$

When $f \in (1, 1.25)$ kHz, $1 = D_1(f)\,Q_1(f) + D_1(f-2)\,Q_1(f-2) \Rightarrow Q_1(f) = \frac{1}{1.5-|f|}$

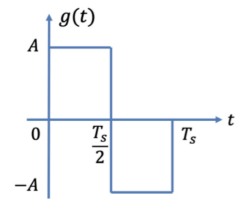
$= Q_1(f)\,[1.5 - |f|]$

d) $$H(f) = \begin{cases} 1, & |f| \in [0, 0.75]\text{ kHz} \\ 0, & |f| \in (0.75, +\infty)\text{ kHz} \end{cases}$$

Bandwidth $W = 0.75\,kHz$, $R_s = 2kHz > 2W = 1.5\,kHz$

$\Rightarrow$ no-ISI design isn't existed

4. 4. In a baseband transmission system, the output waveform of the transmitter is given below. The channel is addictive white Gaussian noise (AWGN) channel with noise PSD $n_0/2$.

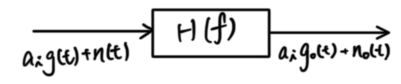


a) Design a matched filter for the optimal receiver and draw the impulse response of the filter.

b) Draw the output of the matched filter.

c) Compute the optimal SNR at the sample output after the matched filter.

a)



$$E[n_o^2(T_s)] = R_{N_o}(0) = \int_{-\infty}^{+\infty} S_{N_o}(f)\, e^{i\cdot 2\pi f 0}\, df = \frac{n_o}{2}\int_{-\infty}^{+\infty}|H(f)|^2\, df$$

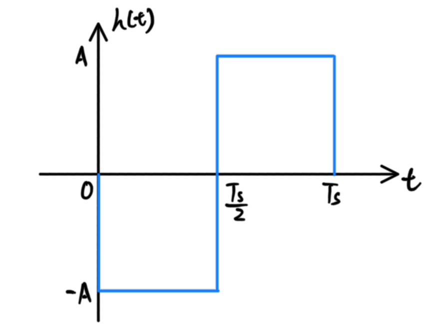
$$|a_i g_o(T_s)|^2 = \left|\int_{-\infty}^{+\infty} a_i\, G(f)\, H(f)\, e^{i2\pi f T_s}\, df\right|^2$$

$$SNR = \frac{|a_i g_o(T_s)|^2}{E[n_o^2(T_s)]} = \frac{2|a_i|^2}{n_o}\frac{\left|\int_{-\infty}^{+\infty} G(f)\, H(f)\, e^{i\cdot 2\pi f T_s}\, df\right|^2}{\int_{-\infty}^{+\infty}|H(f)|^2\, df}$$

$$\le \frac{2|a_i|^2}{n_o}\int_{-\infty}^{+\infty}|G(f)|^2\, df \quad \left(\text{The equality holds iff } H(f) = \lambda G^*(f)\, e^{-i\cdot 2\pi f T_s}\right)$$

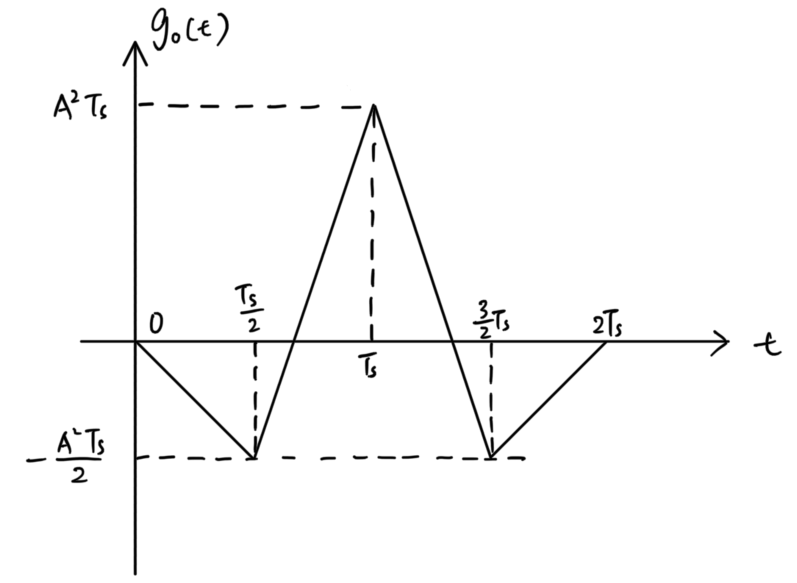
Match filter : $H(f) = G^*(f)\, e^{-i\cdot 2\pi f T_s}$

$h(t) = g(T_s - t)$



b) $$g_o(t) = \int_{-\infty}^{+\infty} g(\tau)\, h(t-\tau)\, d\tau = \int_{-\infty}^{+\infty} g(\tau)\, g(\tau + T_s - t)\, d\tau$$

$$= \begin{cases} -A^2 t, & 0 \le t \le \frac{T_s}{2} \\ -A^2(2T_s - 3t), & \frac{T_s}{2} < t \le T_s \\ A^2(4T_s - 3t), & T_s < t \le \frac{3}{2}T_s \\ -A^2(2T_s - t), & \frac{3}{2}T_s < t \le 2T_s \\ 0, & t \notin [0, 2T_s] \end{cases}$$



c) $$SNR = \frac{2|a_i|^2}{n_o}\int_{-\infty}^{+\infty}|G(f)|^2\, df = \frac{2|a_i|^2}{n_o}\int_{-\infty}^{+\infty}|g(t)|^2\, dt = \frac{2|a_i|^2}{n_o} A^2 T_s$$

(Perseval Equality)

5. 5. In this problem, we derive the matched filter in the time domain.

Suppose a symbol a_i is modulated on a baseband pulse waveform $g(t)$. The receiver filter is $h(t)$ and is sampled at $t = T_s$. Consider an additive white Gaussian noise $n(t)$ with PSD (Power Spectral Density) $n_0/2$. The sampled signal is:

$$Y = S + N,$$

where $S = a_i \int_{-\infty}^{\infty} g(\tau)h(T_s - \tau)\, d\tau$, $N = \int_{-\infty}^{\infty} n(\tau)h(T_s - \tau)\, d\tau$.

a) Prove the expression of the Signal-to-Noise Ratio (SNR) at the output sample point is

$$\text{SNR} = \frac{S^2}{\mathbb{E}[N^2]} = \frac{2a_i^2}{n_0}\frac{\left[\int_{-\infty}^{\infty} g(\tau)h(T_s - \tau)\, d\tau\right]^2}{\int_{-\infty}^{\infty} h^2(\tau)\, d\tau}$$

b) Find the maximum SNR and the corresponding $h(t)$.

(Hint: use the Cauchy-Schwarz inequality: $(\int f(x)g(x)\, dx)^2 \le \int f^2(x)\, dx \int g^2(x)\, dx$, with equality if and only if $g(x) = kf(x)$)

a) $$|S|^2 = |a_i|^2\left|\int_{-\infty}^{+\infty} g(\tau)\, h(T_s - \tau)\, d\tau\right|^2$$

$$E[|N|^2] = E\left[\left(\int_{-\infty}^{+\infty} n(\tau)\, h(T_s - \tau)\right)\left(\int_{-\infty}^{+\infty} n^*(\tau')\, h^*(T_s - \tau')\, d\tau'\right)\right]$$

$$= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} h(T_s - \tau)\, h^*(T_s - \tau')\, E[n(\tau)\, n^*(\tau')]\, d\tau\, d\tau'$$

$$= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} h(T_s - \tau)\, h^*(T_s - \tau')\cdot\frac{n_o}{2}\delta(\tau' - \tau)\, d\tau\, d\tau'$$

$$= \frac{n_o}{2}\int_{-\infty}^{+\infty} h(T_s - \tau)\, h^*(T_s - \tau)\, d\tau$$

$$= \frac{n_o}{2}\int_{-\infty}^{+\infty}|h(T_s - \tau)|^2\, d\tau$$

$$= \frac{n_o}{2}\int_{-\infty}^{+\infty}|h(\tau)|^2\, d\tau$$

$$\Rightarrow SNR = \frac{|S|^2}{E[|N|^2]} = \frac{2|a_i|^2}{n_o}\frac{\left|\int_{-\infty}^{+\infty} g(\tau)\, h(T_s - \tau)\, d\tau\right|^2}{\int_{-\infty}^{+\infty}|h(\tau)|^2\, d\tau}$$

b) $$\left|\int_{-\infty}^{+\infty} g(\tau)\, h(T_s - \tau)\, d\tau\right|^2 \le \int_{-\infty}^{+\infty}|g(\tau)\, h(T_s - \tau)|^2\, d\tau \le \left(\int_{-\infty}^{+\infty}|g(\tau)|^2\, d\tau\right)\left(\int_{-\infty}^{+\infty}|h(T_s - \tau)|^2\, d\tau\right) = \left(\int_{-\infty}^{+\infty}|g(\tau)|^2\, d\tau\right)\left(\int_{-\infty}^{+\infty}|h(\tau)|^2\, d\tau\right)$$

$$SNR \le \frac{2|a_i|^2}{n_o}\int_{-\infty}^{+\infty}|g(\tau)|^2\, d\tau \quad [\text{Equality holds iff } h(\tau) = \lambda g^*(T_s - \tau),\ \lambda \text{ is a constant}]$$

$$\text{Maximum } SNR = \frac{2|a_i|^2}{n_o}\int_{-\infty}^{+\infty}|g(\tau)|^2\, d\tau$$

Corresponding $h(t) = \lambda g^*(T_s - t)$